

VEER PAWAR

AI Engineer · RAG Pipelines · Agentic AI · Computer Vision

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Open to Pan-India Relocation | SIH 2025 National Finalist | International Agri Hackathon Runner-Up | 2× National Hackathon Finalist



ABOUT ME

Final-year AI & Data Science student (graduating 2026, CGPA 8.0/10) with real production experience — not just classroom projects. During my internship at Cloudzen, I designed and shipped an end-to-end agentic OCR + FAISS + LLM pipeline that cut manual procurement effort by around 40%. I'm comfortable across the full stack: from raw data ingestion and model training to REST API deployment and cross-team delivery. Looking for an AI/ML engineering role where the work actually goes live and makes a measurable difference.

TECHNICAL SKILLS

Core AI / ML	Python, TensorFlow, PyTorch, Scikit-learn, YOLOv8, CNN, Transfer Learning, VGG16
LLM & RAG	LLM Integration, RAG Pipelines, FAISS, Sentence Embeddings, Prompt Engineering, OCR, Agentic AI
Data & APIs	FastAPI, Flask, REST APIs, ETL Pipelines, SQL, PySpark, Data Modeling, Web Scraping
Cloud & DevOps	AWS
Viz & Analytics	Power BI, Pandas, NumPy, Matplotlib

PROFESSIONAL EXPERIENCE

Software Engineer Intern

Cloudzen Innovations Pvt. Ltd.

Dec 2025 – July 2026

AI Voice Agent — Transport Procurement Platform Python · LiveKit · FastAPI · Sarvam STT/TTS · LLM · Sentence Transformers · Redis · RAG

- Built a real-time AI voice agent that autonomously calls freight suppliers, negotiates transport rates across target/yellow/red threshold zones, and manages the full call lifecycle — from identity confirmation to rate closure — over a live phone line.
- Designed a 3-layer intent detection pipeline: deterministic rule-matching runs first, followed by multilingual embedding similarity (intfloat/multilingual-e5-small with cosine scoring), falling back to LLM classification — achieving reliable intent recognition across Hindi, Marathi, and English supplier responses.
- Implemented a lightweight RAG system that chunks and indexes logistics JSON/text data at startup, scores chunks via token-overlap ranking, and injects relevant context into the LLM prompt at each turn to ground responses in actual business data.
- Integrated Sarvam STT (saarika:v2.5) and TTS (bulbul:v3) with per-language voice, pace, and pitch tuning across 11 Indian language codes, and wired the full pipeline into a FastAPI + LiveKit server with Redis-backed call-task queuing.

KEY PROJECTS

AI-Powered Garbage & Spills Detection

YOLOv8 · Computer Vision · Python · CCTV · Task Scheduling

- Fine-tuned YOLOv8 on 2,000+ self-annotated images and deployed real-time detection on live CCTV infrastructure — hitting **24 FPS** throughput with **sub-2-second** automated crew dispatch alerts. Built for facilities where missing an incident isn't an option.

Student Engagement Tracking via Facial Cues

VGG16 · CNN · TensorFlow · Transfer Learning · Flask

- Trained VGG16 on the FER-2013 dataset and reached **87%+** emotion classification accuracy across seven classes. Served via a Flask REST API at **under 100 ms** latency — runs on standard hardware without a GPU.

Text Sentiment Analysis — NLP Classification Pipeline

Python · NLP · Scikit-learn · TF-IDF · SQL · Power BI

- Hit **91%+** binary classification accuracy on large-scale review data using TF-IDF + Logistic Regression and Random Forest. End-to-end pipeline from raw data ingestion through model evaluation to a live Power BI dashboard.

AI Transport Procurement Agent

Python · LLM · Agentic AI · FAISS · RAG · OCR

- Processed **500+** vendor documents per month through OCR extraction, sentence embeddings, and FAISS vector search — cut manual vendor scoring time by **~40%** via structured LLM output parsing and context-aware retrieval.

Automated Price Tracker — Multi-Platform ETL

Python · ETL · Web Scraping · Automation · Email Alerts

- Tracked **300+ SKUs** across four platforms at **99% uptime** over three months. Scheduled ETL with deduplication, schema validation, and event-triggered email alerts on configurable price-drop thresholds.

Single Object Tracking System

BoT-SORT · Python · Video Analytics · Sequential Data

- Real-time object tracking at **30 FPS** in dynamic video environments using BoT-SORT, with MOTA/MOTP evaluation metrics and temporal data modelling for sequential stream processing.

EDUCATION

B.Tech — Artificial Intelligence & Data Science CSMSS CHH Shahu College of Engineering, Aurangabad	CGPA 8.0 / 10 · 2022 – 2026
HSC — Science R.K. High School & Jr. College	78% · 2022
SSC — CBSE G.C.P.S. Pulgaon	82% · 2020

Relevant Coursework: Machine Learning · Deep Learning · NLP · Computer Vision · Data Analytics · Database Management · Data Engineering

CERTIFICATIONS & RECOGNITION

- Machine Learning, Deep Learning & Data Analytics Using Python — NPTEL (Verified)
- Python Programming — HackerRank (Certified)
- SQL — HackerRank (Certified)